

Reduction Formulas

Sine:

$$\int \sin^n x \, dx = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$

cos x:

$$\int \cos^n x \, dx = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} \int \cos^{n-2} x \, dx$$

tan x:

$$\int \tan^n x \, dx = \frac{\tan^{n-1} x}{n-1} - \int \tan^{n-2} x \, dx$$

cot x:

$$\int \cot^n x \, dx = -\frac{\cot^{n-1} x}{n-1} - \int \cot^{n-2} x \, dx$$

sec x:

$$\int \sec^n x \, dx = \frac{\tan x \sec^{n-2} x}{n-1} + \frac{n-2}{n-1} \int \sec^{n-2} x \, dx$$

cosec x:

$$\int \operatorname{cosec}^n x \, dx = \frac{-\cot x \operatorname{cosec}^{n-2} x}{n-1} + \frac{n-2}{n-1} \int \operatorname{cosec}^{n-2} x \, dx$$

Integration Formulas:

$$\cdot \int x^n \, dx = \frac{x^{n+1}}{n+1} + c$$

$$\cdot \int \cot x \, dx = \ln |\sin x| + c$$

$$\cdot \int 1 \, dx = x + c$$

$$\cdot \int \operatorname{cosec} x = \ln |\operatorname{cosec} x - \cot x| + c$$

$$\cdot \int \cos x \, dx = \sin x + c$$

$$\cdot \int \sec x = \ln |\sec x + \tan x| + c$$

$$\cdot \int \sin x \, dx = -\cos x + c$$

$$\cdot \int \tan x \, dx = \ln |\sec x| + c$$

$$= -\ln |\cos x| + c$$

$$e) \int \sin^4 x \, dx$$

$$\begin{aligned}
 n &= 4 & -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} \int \sin^{n-2} x \, dx \\
 &= -\frac{\sin^3 x \cos x}{4} + \frac{3}{4} \int \sin^2 x \, dx \\
 &= -\frac{\sin^3 x \cos x}{4} + \frac{3}{4} \left[-\frac{\sin x \cos x}{2} + \frac{1}{2} \int 1 \, dx \right] \\
 &= -\frac{\sin^3 x \cos x}{4} + \frac{3}{4} \left[-\frac{\sin x \cos x}{2} + \frac{1}{2} x \right] \\
 &= -\frac{\sin^3 x \cos x}{4} - \frac{3 \sin x \cos x}{8} - \frac{3x}{8} + C
 \end{aligned}$$

$$e) \int \sin^7 x \, dx$$

$$\begin{aligned}
 n &= 7 & -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} \int \sin^{n-2} x \, dx \\
 &= -\frac{\sin^6 x \cos x}{7} + \frac{6}{7} \int \sin^6 x \, dx \\
 &= -\frac{\sin^6 x \cos x}{7} + \frac{6}{7} \left[-\frac{\sin^5 x \cos x}{5} + \frac{4}{5} \int \sin^4 x \, dx \right] \\
 &= -\frac{\sin^6 x \cos x}{7} + \frac{6}{7} \left[-\frac{\sin^5 x \cos x}{5} + \frac{4}{5} \left[-\frac{\sin^3 x \cos x}{3} + \frac{2}{3} \int \sin^2 x \, dx \right] \right] \\
 &= -\frac{\sin^6 x \cos x}{7} + \frac{6}{7} \left[-\frac{\sin^5 x \cos x}{5} + \frac{4}{5} \left[-\frac{\sin^3 x \cos x}{3} - \frac{2}{3} \cos x \right] \right] + C \\
 &= -\frac{\sin^6 x \cos x}{7} + \frac{6}{7} \left[-\frac{\sin^5 x \cos x}{5} - \frac{4 \sin^3 x \cos x}{15} - \frac{8 \cos x}{15} \right] + C \\
 &= -\frac{\sin^6 x \cos x}{7} - \frac{6 \sin^5 x \cos x}{35} - \frac{24 \sin^3 x \cos x}{105} - \frac{48 \cos x}{105} + C
 \end{aligned}$$

$$e) \int \cos^7 x \, dx$$

$$n = 7$$

$$= \frac{\cos^6 x \sin x}{7} + \frac{6}{35} \cos^4 x \sin x + \frac{8}{35} \cos^2 x \sin x + \frac{16}{35} \sin x + C$$

$$Q) \int \cos^6 x \, dx$$

$$n = 6$$

$$= \frac{\cos^5 x \sin x}{5} + \frac{5}{24} \cos^3 x \sin x + \frac{3}{16} \cos x \sin x + \frac{3}{16} x + C$$

$$Q) \int \cot^3 x \, dx$$

$$= -\frac{\cot^6 x}{6} + \frac{\cot^4 x}{4} - \frac{\cot^2 x}{2} - \ln |\sin x| + C$$

$$Q) \int \tan^5 x \, dx$$

$$= \frac{\tan^4 x}{4} - \frac{\tan^2 x}{2} + \ln |\sec x| + C$$

$$Q) \int \sec^5 x \, dx$$

$$= \frac{\tan x \sec^3 x}{4} + \frac{3}{8} \tan x \sec x + \frac{1}{8} \ln |\sec x + \tan x| + C$$

$$Q) \int \operatorname{cosec}^3 x \, dx$$

$$= -\frac{\cot x \operatorname{cosec}^5 x}{6} - \frac{5}{24} \cot x \operatorname{cosec}^3 x - \frac{5}{16} \cot x \operatorname{cosec} x + \frac{5}{16} \ln |\operatorname{cosec} x - \cot x| + C$$

